Leander Paul

Senior Backend Engineer — Identity & Banking Platforms

Dubai, UAE

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Summary

Senior backend engineer with 8+ years of experience, most recently building banking and identity systems in Dubai. At Emirates NBD (via Sphere IT), architected an authorization microservice supporting OTP, IVR and Nafath with configurable mid-flow validation, built the manual-review stage of the in-app credit card application, and own escalated production issues and the disaster-recovery environment setup. Previously worked on Zoho Accounts IAM (SAML setup, hierarchical roles and permissions). Independently built Shadow Platform, a public TypeScript monorepo including an OIDC/OAuth 2.1 identity provider, developed with AI-assisted tooling; since 2026 also building LLM applications.

Experience

Software Engineer — Sphere IT Consultants

Client: Emirates NBD · Dubai, United Arab Emirates · Mar 2024 – Present

- Architected a configurable authorization microservice that authenticates customers via OTP, IVR and Nafath (Saudi national identity), running database-configured validation checks, such as device trust, between steps and cancelling the flow when a check fails.
- Built the manual-review stage for partially eligible credit card applicants and contributed to the in-app application journey, integrating external credit-scoring and employment-verification services.
- Own production issues that application support cannot resolve: reproduce, find the root cause and ship fixes to live banking services; responsible for setting up the disaster-recovery environment.

Technologies: Java, Spring Boot, Node.js, Express

Career break · Oct 2023 – Feb 2024: Relocated to Dubai; rebuilt my Shadow Accounts identity service (NestJS/Fastify) and a shared service library (public on GitHub).

Full Stack Engineer (Contract) — WTV Global

London, United Kingdom · Mar 2023 – Sep 2023

- Helped design the architecture and APIs of Capture, a carbon-footprint platform built as three NestJS microservices over RabbitMQ (auth and users; projects and permissions; calculations), and developed its backend services.
- Initiated and set up end-to-end tests that reset and reseed the database before each run; services were deployed as Docker containers.

Technologies: NestJS, RabbitMQ, Docker

Software Engineer — Zoho Corporation

Chennai, India · Jan 2020 – Aug 2022

- Designed a hierarchical role-and-permission model for Zoho Accounts IAM and worked on SAML setup.
- Built a proof of concept that replaced Apache Directory Server's storage layer with the team's PostgreSQL database, so customers could query it over LDAP.
- Helped identify Redis server crashes caused by CPU load from TTL handling and provided a general solution that balanced the load without crashes.
- Wrote migration scripts for application-level sharded databases and set up and optimised the admin panel the accounts team uses to inspect and clear user caches; helped mentor junior engineers.

Technologies: Java, JavaScript, PostgreSQL, Redis, Apache Directory Server, SAML, LDAP

Full Stack Engineer & Team Lead — Phosphene AI

Chennai, India · Apr 2018 – Jan 2020

- Designed and built PixelView's upload-to-upscale pipeline: browser-direct S3 uploads, Lambda notifications, an SQS queue, and an upscaler server started on demand and stopped after 5 idle minutes to cut compute cost.

- Led a team of 3 developers and wrote most of the frontend (React, Next.js) and Node.js backend, including Google sign-in, Google Drive import/export and Stripe payments.

Technologies: Node.js, React, Next.js, Python, MongoDB, AWS (EC2, S3, Lambda, SQS), Stripe

Software Engineer (Freelance)
Chennai, India · Apr 2017 – Mar 2018

- Built and deployed two web applications on AWS EC2: Bluebell, a Node.js and Handlebars site with form submissions, and Jute Emporium, an e-commerce store with Razorpay payments, an admin panel for stock and orders, and order-status emails.

Technologies: Node.js, Handlebars, MongoDB, AWS EC2

Projects

Shadow Platform — personal project, public
github.com/shadow-library/platform · 2024 – Present

- Built a public TypeScript monorepo of six products (identity, notifications, AI gateway, novel authoring, reading, life tracking), each a server and web app, with servers on my own dependency-injection framework; 10,000+ automated tests (unit, integration, e2e); developed solo with AI-assisted tooling.
- Identity: OIDC/OAuth 2.1 provider (PKCE-only, refresh rotation with reuse detection), RBAC decision point, M2M and workload identity, SAML 2.0 IdP, passkeys, TOTP, back-channel logout, RFC 8693 delegation (single-hop).
- AI platform: LangGraph workflows checkpointed in PostgreSQL with idempotent retry and a generate → judge → repair loop; model calls routed through Synapse, an internal gateway with a model catalog, failover, budgets and a usage ledger.
- Self-hosted development environment using k3d/GitOps; production deployment planned.

Open source: @shadow-library npm packages (app, common, class-schema, fastify, modules; MIT), now maintained in the platform monorepo.

Education

MSc Computer Science — Queen Mary University of London — 2022 – 2023
Bachelor of Technology — St. Joseph's College of Engineering (Anna University), Chennai — 2016 – 2020

Skills

Backend: Java, Spring Boot, Node.js, TypeScript, Express, NestJS, Fastify, Bun, REST APIs, microservices
Identity & security: OAuth 2.0/2.1, OpenID Connect, SAML 2.0, WebAuthn/passkeys, TOTP MFA, RBAC, LDAP, OTP/IVR/Nafath authentication
Data & messaging: PostgreSQL, Drizzle ORM, Redis, MongoDB, RabbitMQ, AWS SQS
Cloud & delivery: AWS (EC2, S3, Lambda, SQS), Docker, Kubernetes (k3d), GitOps, CI/CD (GitHub Actions)
Applied AI (personal projects): LangGraph, LangChain, pgvector, structured output, LLM evaluation, MCP
Frontend: React, Next.js, Vue, TanStack Start
Practices: system design, production incident analysis, disaster recovery, end-to-end testing, mentoring